

PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE
NAME

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CT
GROUP

1	4			
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INDEX
NUMBER

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CHEMISTRY

9647/02

Paper 2 Structured

15 September 2015

2 hours

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your CT group, index number and name on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

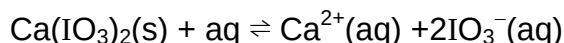
FOR EXAMINER'S USE			
Paper 2			
1	/ 12	5	/ 12
2	/ 7	6	/ 15
3	/ 11	Penalty	sf units
4	/ 15	Total	/ 72

Answer **all** the questions in the spaces provided.

1 Planning (P)

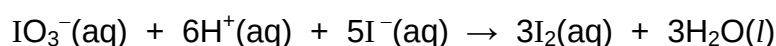
(a) The solubility of calcium iodate(V), $\text{Ca}(\text{IO}_3)_2$, is approximately 5 g dm^{-3} .

When solid calcium iodate(V) is added to water, a small amount dissolves to form a saturated solution, establishing an equilibrium between the solid and its aqueous ions.

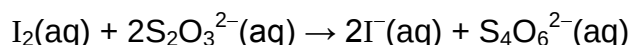


You are given a solid, **FA1**, containing $\text{Ca}(\text{IO}_3)_2$. You are asked to determine the exact concentration of IO_3^{-} ions in a saturated solution of $\text{Ca}(\text{IO}_3)_2$, and hence the solubility product of $\text{Ca}(\text{IO}_3)_2$, by carrying out two reactions:

- reaction of acidified IO_3^{-} with I^{-} to liberate aqueous I_2 ,



- the titration of the I_2 liberated against $\text{Na}_2\text{S}_2\text{O}_3$.



(i) Using the information provided, you are required to write a plan to determine as accurately as possible, the solubility product of calcium iodate(V).

You are provided with the following:

- **FA1**: solid calcium iodate(V), $\text{Ca}(\text{IO}_3)_2$
- **FA2**: aqueous potassium iodide, KI
- **FA3**: aqueous sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$
- **FA4**: aqueous hydrochloric acid, HCl
- starch indicator
- apparatus normally found in a college laboratory

Your plan should include:

- brief, but specific details of the apparatus you would use, bearing in mind the levels of precision they offer,
- details, including quantities for the preparation of the saturated solution of $\text{Ca}(\text{IO}_3)_2$ from solid **FA1**,
- essential details of the titration procedure.

[5]

Preparation of saturated solution of calcium iodate for titration

Note: Mass of solid $\text{Ca}(\text{IO}_3)_2$ ($> 0.5\text{g}$) dissolved in 100 cm^3

- Using a weighing balance, weigh out 0.700 g solid $\text{Ca}(\text{IO}_3)_2$.
- Add 0.700 g solid $\text{Ca}(\text{IO}_3)_2$ to 100 cm^3 of distilled water in a

250 cm³ beaker, stir until no more solid dissolves.

-
- Let it stand for 15 minutes to allow equilibrium to be reached.
- Using a filter funnel and filter paper, filter and collect the filtrate which contains saturated Ca(IO₃)₂ solution.

Titration

- Pipette 25.0 cm³ of this saturated solution into a 250 cm³ conical flask
- Using 10 cm³ measuring cylinder, add 5 cm³ KI (excess) to the flask. Use another 10 cm³ measuring cylinder, add about 5 cm³ of HCl (aq) to the flask.
- Swirl the flask to ensure even mixing.
- Titrate the saturated solution with Na₂S₂O₃ from the 50 cm³ burette into the flask
- When the brown colour of the iodine fades to a pale yellow colour, add about 3 drops of starch solution to the flask.
- Continue adding Na₂S₂O₃ until the blue-black colour just disappears.
- Repeat the titrations to obtain at least two consistent results within ± 0.10 cm³.

- (ii) A student found that 25.0 cm³ of the saturated solution of Ca(IO₃)₂, upon treatment with excess **FA2** and **FA4**, requires v dm³ of **M** mol dm⁻³ **FA3** for complete reaction.

Show how you would use her results to determine the solubility product of Ca(IO₃)₂ in terms of v and **M**. [3]

v = volume of S₂O₃²⁻ required for titration (dm³)

$n(\text{S}_2\text{O}_3^{2-})$ required for titration = $v \times \text{concentration of S}_2\text{O}_3^{2-}$
 $= v \times \mathbf{M}$

1 mol IO₃⁻ $\equiv$ 3 mol I₂ $\equiv$ 6 mol S₂O₃²⁻

$\text{Ca}(\text{IO}_3)_2(\text{s}) + \text{aq} \rightleftharpoons \text{Ca}^{2+}(\text{aq}) + 2\text{IO}_3^{-}(\text{aq})$

$[\text{IO}_3^{-}] = \frac{v\mathbf{M}}{6(25 \times 10^{-3})} \text{ mol dm}^{-3}$

$[\text{Ca}^{2+}] = \frac{1}{2} \left[\frac{v\mathbf{M}}{6(25 \times 10^{-3})} \right] \text{ mol dm}^{-3}$

$K_{\text{sp}} = [\text{Ca}^{2+}] [\text{IO}_3^{-}]^2 = \frac{1}{2} \left[\frac{v\mathbf{M}}{6(25 \times 10^{-3})} \right]^3$

(b) **FA5** is a solution containing a mixture of bromide and iodide ions.

You are provided with the following reagents and laboratory apparatus:

- HCl(aq)
- concentrated $\text{NH}_3(\text{aq})$
- dilute $\text{NH}_3(\text{aq})$
- $\text{AgNO}_3(\text{aq})$
- $\text{HNO}_3(\text{aq})$
- deionised water
- test tubes
- filter funnel
- filter paper

Using **only the reagents and materials provided**, devise a sequence of steps by which the two anions could be separately obtained as precipitates.

In your plan you should:

give a full description of your proposed sequence of steps, numbering each step;

state the location of each anion (e.g. in a solution or precipitate) after each step.

[4]

	Procedure	Location of anion
1	To 1 cm ³ of the mixture FA5 in a test tube, add $\text{AgNO}_3(\text{aq})$ dropwise, until in excess to the solution.	Both Br^- and I^- are present as precipitates.
2	Add excess concentrated $\text{NH}_3(\text{aq})$ dropwise (AgBr dissolves in excess concentrated NH_3 to give a colourless solution of $\text{Ag}(\text{NH}_3)_2^+\text{Br}^-$ while AgI remains insoluble)	Br^- ion is in solution. I^- remains as a precipitate.
3	Filter the mixture and wash the residue with distilled water. Dry the residue by pressing between filter papers.	Br^- is in the filtrate. I^- is in the residue.
4	To the filtrate from above, add dropwise of $\text{HNO}_3(\text{aq})$ in excess. Filter and wash the residue with distilled water. Dry the residue by pressing between filter papers.	Br^- is in precipitate.

[Total: 12]

- 2 Stink bombs can be constructed using solid ammonium hydrosulfide, which decomposes to form ammonia and hydrogen sulfide readily.



- (a) Calculate the average M_r of the gaseous mixture.

$$M_r \text{ of NH}_3 = 17$$

$$M_r \text{ of H}_2\text{S} = 34.1$$

$$\text{Average } M_r = (34.1 + 17) / 2 = 25.55 = 25.6$$

[1]

- (b) When 0.500 g of solid NH_4SH was placed in a 350 cm^3 sealed flask at 20°C and left to equilibrate for a day, 0.265 g of solid was found to remain.

Using your result from (a), calculate the total pressure of gas in the sealed flask.

$$p_{\text{total}}V = n_{\text{total}}RT$$

$$\begin{aligned} \text{total amount of gases at eqm, } n_{\text{total}} &= n_{\text{H}_2\text{S}} + n_{\text{NH}_3} \\ &= (0.500 - 0.265) / 25.6 \\ &= 9.198 \times 10^{-3} \text{ mol} \end{aligned}$$

$$\begin{aligned} p(350 \times 10^{-6}) &= (9.198 \times 10^{-3}) \times (8.31)(20 + 273) \\ p &= 63985 \text{ Pa} = 6.40 \times 10^4 \text{ Pa} \end{aligned}$$

[2]

- (c) (i) Write an expression for K_p of the decomposition of ammonium hydrosulfide.

[1]

$$K_p = (P_{\text{H}_2\text{S}})(P_{\text{NH}_3})$$

- (ii) Using your answer in (b), determine the K_p for the decomposition of ammonium hydrosulfide.

$$P_{\text{NH}_3} = 6.40 \times 10^4 / 2 = 3.20 \times 10^4 \text{ Pa}$$

$$K_p = (3.20 \times 10^4)^2 = 1.02 \times 10^9 \text{ Pa}^2$$

[2]

- (iii) Explain how the position of equilibrium would be affected when more solid NH_4SH is added to the equilibrium mixture.

The addition of solid NH_4SH does not change the partial pressure of NH_3 and H_2S , hence no impact on the value of K_p . As such the position of equilibrium would remain unchanged.

[Total: 7]

3 (a) The solubility of $\text{Mg}(\text{OH})_2$ in water at 25°C is $1.20 \times 10^{-2} \text{ g dm}^{-3}$.

(i) Calculate the value of K_{sp} of $\text{Mg}(\text{OH})_2$, stating its units. [2]

$$\begin{aligned}\text{Solubility of } \text{Mg}(\text{OH})_2 &= 1.20 \times 10^{-2} / 58.3 \\ &= 2.058 \times 10^{-4} \text{ mol dm}^{-3}\end{aligned}$$

$$\begin{aligned}K_{\text{sp}} &= (2.058 \times 10^{-4})(2 \times 2.058 \times 10^{-4})^2 \\ &= 3.49 \times 10^{-11} \text{ mol}^3 \text{ dm}^{-9}\end{aligned}$$

(ii) Predict whether a precipitate of $\text{Mg}(\text{OH})_2$ would be observed, when equal volumes of a $0.10 \text{ mol dm}^{-3} \text{ Mg}^{2+}(\text{aq})$ solution and a buffer solution at pH 9.5 are mixed. [2]

$$\begin{aligned}\text{pOH} &= 14 - 9.5 \\ &= 4.5\end{aligned}$$

$$[\text{OH}^-] = 10^{-4.5} \text{ mol dm}^{-3}$$

$$\text{Ionic product} = (0.05)(10^{-4.5})^2 = 5.00 \times 10^{-11} \geq K_{\text{sp}}$$

Note: as pH of buffer remains unchanged on dilution, $[\text{OH}^-]$ should not change on mixing.

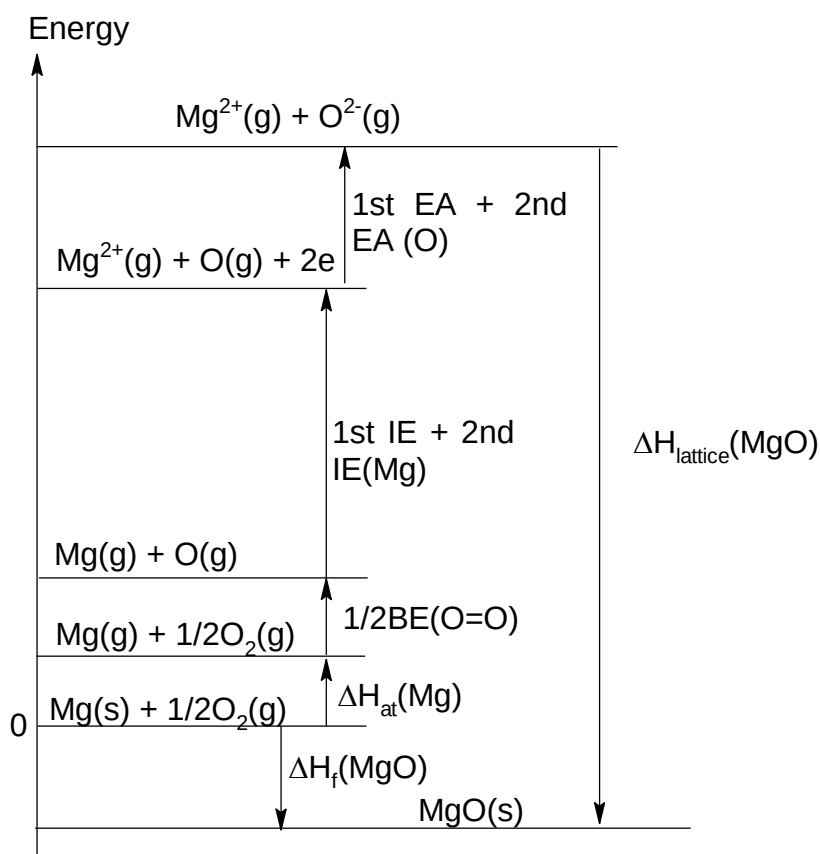
A precipitate will be observed.

(b) $\text{Mg}(\text{OH})_2$ undergoes decomposition on heating to give the corresponding oxide. State and explain how the decomposition temperature of $\text{Ca}(\text{OH})_2$ would compare with that of $\text{Mg}(\text{OH})_2$. [2]

Ca^{2+} has a lower charge density than Mg^{2+} due to its larger ionic radius. With a lower polarising power, Ca^{2+} is less able to distort the electron cloud of the OH^- ion, thus the O-H bond is weakened to a smaller extent. A higher temperature is required for $\text{Ca}(\text{OH})_2$ to undergo thermal decomposition.

(c) (i) Construct an energy level diagram using the following data, together with relevant data from the *Data Booklet* to calculate the lattice energy of magnesium oxide. [3]

enthalpy change of atomisation of $\text{Mg}(\text{s})$	$+150 \text{ kJ mol}^{-1}$
first electron affinity of oxygen	-147 kJ mol^{-1}
second electron affinity of oxygen	$+753 \text{ kJ mol}^{-1}$
enthalpy change of formation of $\text{MgO}(\text{s})$	-602 kJ mol^{-1}



[2]

Applying Hess' Law,

$$\Delta H_{\text{lattice}}(\text{MgO}) = -[606 + 736 + 1450 + \frac{1}{2}(496) + 150] - 602$$

$$= -3792 \text{ kJ mol}^{-1} = -3790 \text{ kJ mol}^{-1} \text{ (3 s.f.)}$$

- (ii) Suggest, with reasons, how the magnitude of the lattice energy of $\text{MgF}_2(\text{s})$ might compare to that of $\text{MgO}(\text{s})$. [2]

$$|\Delta H_{\text{lattice}}| \propto \left| \frac{q_+ q_-}{r_+ + r_-} \right|$$

Although the ionic radius of F^- is smaller than that of O^{2-} , O^{2-} has a higher charge. As charge is a more significant factor affecting the lattice energy, the magnitude of lattice energy of MgO is greater than that of MgF_2 .

[Total: 11]

- 4 Copper is a typical *transition element*. It has more than one oxidation state and many of its compounds have colours in the blue-green-yellow part of the visible spectrum.

- (a) What do you understand by the term *transition element*? [1]

A transition element is a d-block element which forms at least one stable ion in which there is a partially filled d-subshell of electrons.

- (b) Explain why the first ionisation energies of transition elements, from titanium to copper, are relatively invariant. [2]

Going across the period from Ti to Cu, the nuclear charge increases due to an increase in number of protons. As the electrons are added to the penultimate 3d orbitals, they shield the valence 4s electrons. The effective nuclear charge, therefore does not increase significantly and the first ionisation energies of the transition elements are invariant.

- (c) What feature of transition elements, such as copper, allows them to show variable oxidation states? [2]

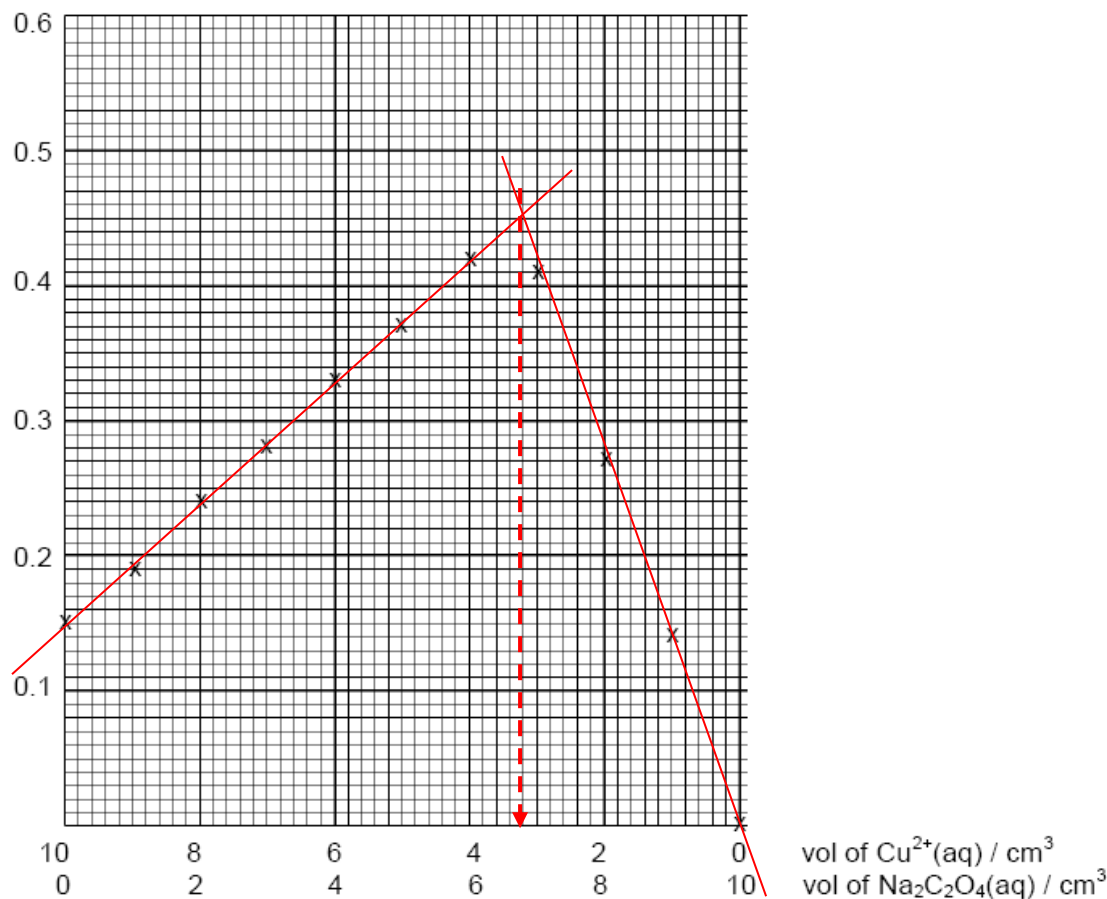
Transition metals have variable oxidation states due to the small energy difference between the 3d and 4s orbitals. Thus, different number of 3d and 4s electrons may be lost / involved in bonding to form stable ions or compounds of different oxidation states and of approximately equal stability.

- (d) Copper forms many complexes with ligands such as H_2O , NH_3 and Cl^- .

To determine the stoichiometry of a copper(II) ethanedioate complex, the colour intensities of solutions containing different proportions of the ligand, $\text{C}_2\text{O}_4^{2-}$, and Cu^{2+} ion were measured.

The intensity of the colour depends on the concentration of the complex ion. The higher the concentration of the complex, the higher is the colour intensity. In one such experiment, the results obtained for different mixtures of $1.0 \times 10^{-4} \text{ mol dm}^{-3}$ $\text{Cu}^{2+}(\text{aq})$ and $1.0 \times 10^{-4} \text{ mol dm}^{-3}$ $\text{Na}_2\text{C}_2\text{O}_4(\text{aq})$ were plotted below.

relative colour intensity



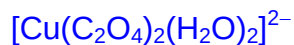
- (i) By drawing two lines of best fit through the plotted points, determine the mole ratio of $\text{Cu}^{2+}(\text{aq})$ to $\text{C}_2\text{O}_4^{2-}(\text{aq})$ in the complex? [2]

The intersection of the two line of best fit gives the proportion of Cu^{2+} and $\text{C}_2\text{O}_4^{2-}$ which gives the highest concentration of the complex.

$$\begin{aligned} n(\text{Cu}^{2+}) : n(\text{C}_2\text{O}_4^{2-}) &= \text{vol of } \text{Cu}^{2+} : \text{vol. of } \text{C}_2\text{O}_4^{2-} \\ &= 3.33 : 6.67 \\ &= 1 : 2 \end{aligned}$$

- (ii) The copper(II) ethanedioate complex has an octahedral shape and it retains some water molecules as ligands.

Suggest a formula for the complex, including its overall charge.



[1]

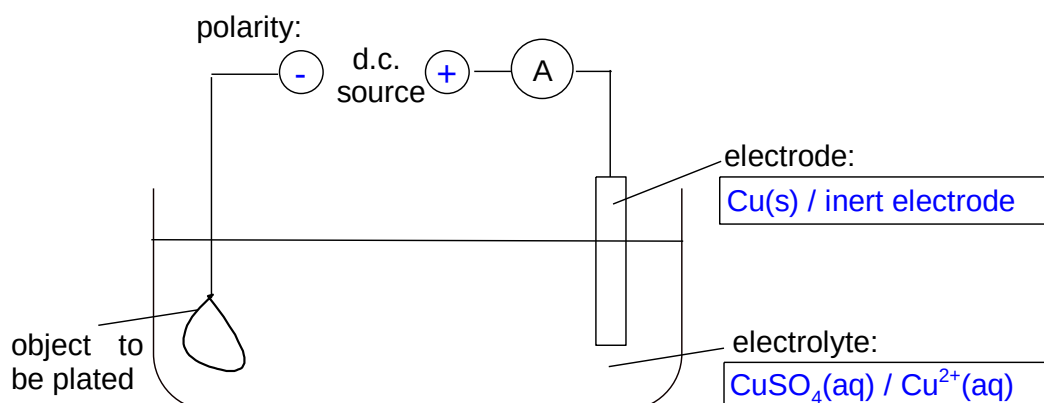
Note: $\text{C}_2\text{O}_4^{2-}$ is a bidentate ligand.

- (e) Explain why complexes of Cu(II) are coloured whereas complexes of Cu(I) are colourless. [3]

For Cu(II) complexes with d^9 configuration or partially filled d-orbitals, the presence of ligands cause the degenerate 3d orbitals of Cu^{2+} to split into 2 sets of different energy (d-orbital splitting). The energy gap corresponds to the energy of the visible region of the electromagnetic radiation. On absorbing energy from the visible light, an electron gets promoted from a lower d-orbital to a half-filled orbital of higher energy (d-to-d transition). The colour of the complex observed is complementary to the colour of the light absorbed. Cu(I) complexes have a fully filled d subshell / d^{10} configuration. Therefore, d-to-d transition cannot take place.

- (f) Because of its increased scarcity, cheaper copper ornaments are no longer made from the solid metal, but from iron that has been copper plated.

- (i) Complete the following diagram showing the set-up for a copper electroplating process. Show clearly the polarity (+/-) of the power source and suggest a suitable electrode and electrolyte to be used. [2]



- (ii) A current of 0.600 A is passed through the electroplating cell. Calculate the time required, in seconds, to deposit a mass of 0.600 g of copper onto the ornament. [2]

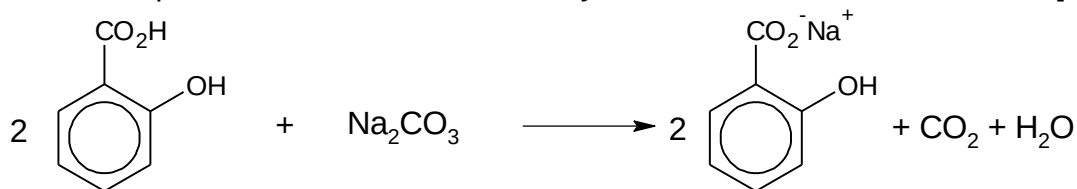
$$\begin{aligned}\text{Amount of Cu to be deposited} &= 0.600 / 63.5 \\ &= 9.45 \times 10^{-3} \text{ mol}\end{aligned}$$

$$\begin{aligned}9.45 \times 10^{-3} &= (0.600 \times t) / (2 \times 96500) \\ t &= 3040 \text{ s}\end{aligned}$$

[Total: 15]

- 5 (a)** Describe a chemical test which can be used to distinguish between compounds **X** and **Y**. State what would be seen with each compound. [2]

- (ii) Write an equation for the reaction of salicylic acid with sodium carbonate. [1]



- (iii) State and explain how the acidity of methyl salicylate compares with that of aspirin. [3]

Aspirin is more acidic than methyl salicylate.

The phenoxide ion formed from methyl salicylate is less stable as the negative charge is only delocalised/ dispersed into the benzene ring.

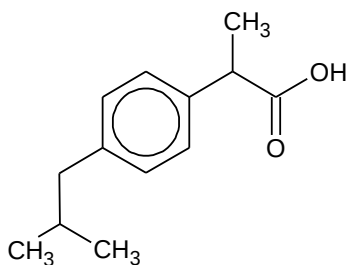
The carboxylate ion formed from the deprotonation of aspirin is more stable than that from methyl salicylate as the negative charge is delocalised/dispersed among 2 electronegative O atoms; hence the negative charge is delocalized/ dispersed to a greater extent than that of phenoxide.

- (iv)** 4-hydroxybenzoic acid, an isomer of salicylic acid, has a melting point of 215 °C while salicylic acid has a melting point of 159 °C.

Suggest an explanation for the difference in the melting points. [2]

In salicylic acid, the $\text{-CO}_2\text{H}$ and -OH in salicylic acid are close to each other, which allows formation of intramolecular hydrogen bonds. This reduces the extent of intermolecular hydrogen bonds to be formed between molecules, thus less energy needed to overcome the forces of attraction between salicylic acid molecules. Hence, salicylic acid has a lower melting point.

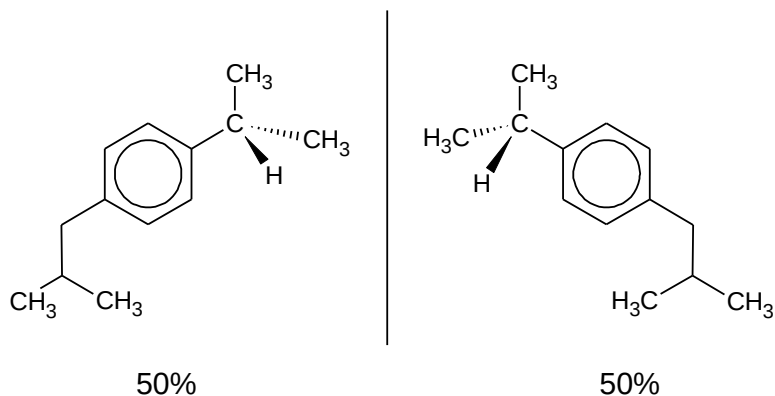
- (c) Ibuprofen has similar anti-inflammatory drug properties as aspirin. A *racemic mixture of ibuprofen* is produced industrially.



Ibuprofen

- (i) With the aid of diagrams, explain what is meant by the term a *racemic mixture of ibuprofen*. [2]

A racemic mixture contains equimolar quantities of the two enantiomers as a result the optical activity cancels out.



- (ii) It was determined that only one enantiomer, the (+)-ibuprofen has anti-inflammatory properties.

An optically pure enantiomer of (+)-ibuprofen has an optical rotation of $+54.5^\circ$. When a racemic mixture of ibuprofen is consumed, an enzyme in the body is able to convert the biologically useless (-)-ibuprofen to (+)-ibuprofen.

A patient consumes a racemic mixture of ibuprofen and a sample of his blood was taken. The ibuprofen in his blood was found to have an optical rotation of $+32.7^\circ$. Determine the percentage of (-)-ibuprofen that was converted to (+)-ibuprofen by the enzymes in the patient's body. [2]

Initial proportion of (+) isomer to (-) isomer = 0.5 : 0.5

Let x be the ratio of (+) isomer after enzyme conversion.

$$(x)(+54.5) + (1-x)(-54.5) = 32.7$$

$$x = 0.8$$

$$\% \text{ of } (-) \text{ ibuprofen converted} = 0.3/0.5 \times 100 = 60\%$$

[Total: 15]

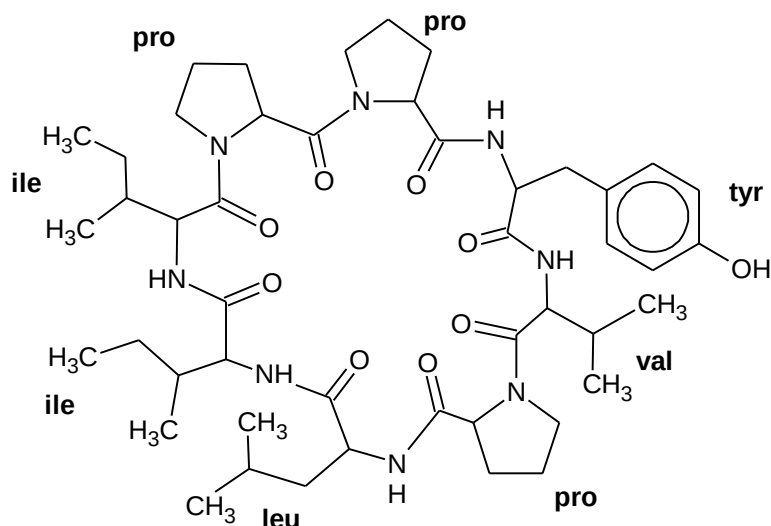
- 6 (a) Suggest, with reasons, how the basicity of phenoxide ion, $\text{C}_6\text{H}_5\text{O}^-$ and ethoxide, $\text{CH}_3\text{CH}_2\text{O}^-$ might compare with that of hydroxide ion, OH^- . [3]

Base strength of $\text{CH}_3\text{CH}_2\text{O}^- > \text{OH}^- > \text{C}_6\text{H}_5\text{O}^-$

$\text{CH}_3\text{CH}_2\text{O}^-$ is a stronger base than OH^- because it has an electron donating alkyl CH_3CH_2- group which increases the electron density of O making its lone pair more available to accept a proton.

$\text{C}_6\text{H}_5\text{O}^-$ is a weaker base than OH^- because the lone pair on O is able to delocalise into the benzene ring. Thus, it is less available to accept a proton.

- (b) Hymenistatin is a cyclic peptide isolated from sea sponges. It is found to have immunosuppressive effects. The structure of hymenistatin is given below along with the abbreviated names of its constituent amino acids: **pro**, **leu** etc.



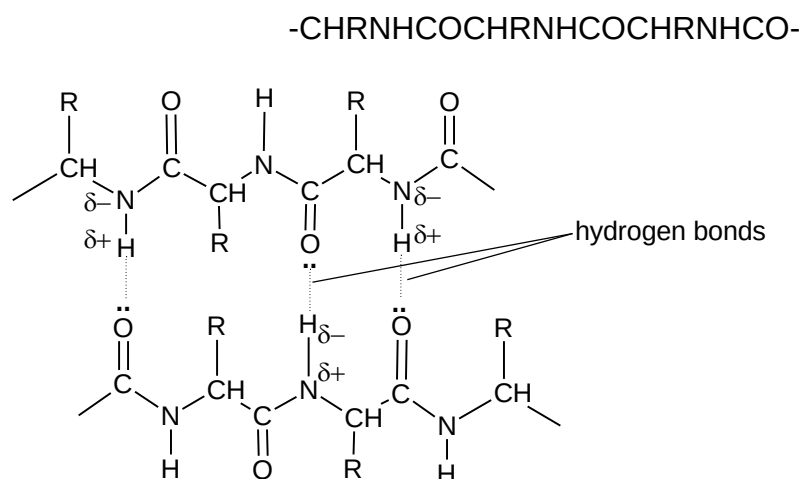
- (i) The enzyme chymotrypsin hydrolyses proteins or polypeptides at the carboxylic end of the amino acid tyrosine, **tyr**.

Use the abbreviated names of the amino acids to state the primary structure of hymenistatin after its hydrolysis by chymotrypsin. [1]

val-pro-leu-ile-ile-pro-pro-tyr

- (ii) Muscle tissues found in meat has a β -pleated sheet structure.

Draw a diagram showing two strands of a β -pleated sheet structure using the formula below. Indicate clearly the interaction between the two strands and show how the R groups are directed. [2]



Both parallel and anti-parallel structures are acceptable.

- (iii) Suggest why the β -pleated sheet structure is unlikely to be found in hymenstatin. [1]

Hymenstatin contains many proline residues which on forming the peptide linkages do not have hydrogen atoms bonded to N available to form hydrogen bond for beta-pleated sheets.

Or

The small ring structure of hymenstatin / the small number of amino acid residues in hymenstatin does not allow the polypeptide to be folded into the beta-pleated sheet.

- (iv) The biological function of hymenstatin depends on its three-dimensional shape.

Suggest, with a reason, the main type of interaction involved in maintaining the three-dimensional shape of hymenstatin. [2]

van der Waals forces are involved in maintaining the three dimensional shape of hymenstatin as the amino acid residues (ile, leu and pro) are mostly non-polar in nature.

- (v) Describe how hymenstatin can be hydrolysed in the laboratory to form a mixture of its constituent amino acids? [1]

Heat with $\text{H}_2\text{SO}_4(\text{aq})$ / $\text{NaOH}(\text{aq})$ under reflux for a long period of time / several hours.

- (vi) Electrophoresis can be used to separate the mixture of amino acids. This technique involves placing the mixture at the centre of an agarose gel placed in an aqueous solution of a suitable pH. An electric potential is then applied through electrodes which are connected to the two ends of the gel strip.

Which pair of amino acids found in hymenstatin is not likely be separated by electrophoresis? Explain your answer. [2]

ile and leu

Both amino acids are isomers / have the same relative molecular mass and they form species with the same charge. Therefore, they would migrate to the same electrode by the same extent when an electric potential is applied.

[Total: 12]

End of Paper